

Slide 1

Disciplines: One Problem, Many Useful Questions

Week 6 Monday — How disciplines shape the
questions we ask

NRES 315: Human Dimensions of Fish and Wildlife Management

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1 / 10

Slide 2

A wolf decision is never only a wolf decision

An agency considering wolf recovery may need to ask about population dynamics. It may also need to ask about fear, livelihoods, identity, authority, and what people owe to one another and to wildlife.

Today: use disciplines as a way to ask better questions before a decision narrows too quickly.

Source: Chapter 7, §7.2

2 / 10

Slide 3

By the end of class, you can

- match a management question to a disciplinary lens;
- distinguish a classic discipline from an applied field; and
- explain when an interdisciplinary or transdisciplinary approach is warranted.

Source: Chapter 7, §7.2

3 / 10

Slide 4

Different lenses make different parts of a problem visible

A discipline is a branch of knowledge that guides inquiry. Each brings characteristic questions, concepts, and ways of developing evidence.

No single discipline owns a conservation problem.

Source: Chapter 7, §§7.2-7.2.1

Slide 5

Four lenses, one wolf-recovery decision

Lens	A question it can make visible
Psychology	Why do people perceive the same risk differently?
Sociology	How do community norms and identities shape support?
Economics	Who bears costs and receives benefits under each option?
Ethics	What responsibilities and harms should guide the choice?

Source: Chapter 7, §§7.2-7.2.4

Slide 6

Psychology asks about minds and behavior

- perceptions of a species, message, or risk;
- beliefs, values, emotions, and intentions; and
- why people respond differently to the same management action.

For example: Why might two residents who receive the same bear-safety message take different actions afterward?

Source: Chapter 7, §§7.2.1.1

Slide 7

Sociology asks about relationships and institutions

- social norms, identities, and community relationships;
- how traditions and opportunities are passed on; and
- how institutions and trust shape conservation outcomes.

For example: How might a hunting tradition be sustained—or interrupted—across generations in one community?

Source: Chapter 7, §7.2.1.2

7 / 10

Slide 8

Economics makes scarcity and distribution explicit

Economics is not simply "putting a dollar value on nature." It asks how limited resources are allocated, what incentives people face, and how costs and benefits are distributed across time and among people.

For example: Who pays for wetland restoration, who benefits, and what changes if landowners face a different incentive?

Source: Chapter 7, §7.2.1.3

8 / 10

Slide 9

Ethics addresses the question evidence cannot settle alone

Data can help estimate likely outcomes. They cannot, by themselves, decide what outcomes ought to matter most.

- Is lethal control justified to protect an endangered species?
- What is owed to people living with a management burden?
- What obligations do present people have to future generations?

Source: Chapter 7, §7.2.1.4

9 / 10

Slide 10

Some fields turn knowledge into action

Applied fields help agencies implement, communicate, regulate, fund, and evaluate decisions in the settings where conservation actually happens.

Source: Chapter 7, §7.2.2

10 / 18

Slide 11

A strong decision needs more than a good explanation

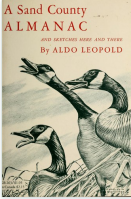
Field	What it contributes to practice
Education	skills, learning, and environmental literacy
Law	authority, constraints, and enforceable rules
Planning	land use, infrastructure, and competing needs
Public administration	accountability, budgets, and coordination
Business	finance, organization, and program delivery

Source: Chapter 7, §7.2.2

11 / 18

Slide 12

Arts and humanities are evidence about meaning not just aesthetics



A Sand County ALMANAC
AND SELECTED ESSAYS AND TALKS
By ALDO LEOPOLD

Stories, images, history, and cultural expression can reveal how people attach meaning to a species or place. They can also communicate consequences in ways a technical report may not.

That does **not** mean substituting a moving image for evidence. It means asking what a solely technical account is unable to hear or convey.

Source: Chapter 7, §7.2.3 and Box 7.1. Cover image: Open Library.

12 / 18

Slide 13

When disciplines meet, the work changes

Complex decisions often require scientists to connect methods—and to work with the people who will live with the decision.

Source: Chapter 7, §7.2.4

13 / 18

Slide 14

Interdisciplinary is not the same as transdisciplinary

- **Multidisciplinary:** several fields address one problem side by side.
- **Interdisciplinary:** fields integrate concepts, theories, or methods.
- **Transdisciplinary:** scientific fields work with practitioners, decision makers, and resource users to define, interpret, and apply the work.

The key question is not "How many disciplines are present?" but "Who helped shape the problem and how are their insights connected?"

Source: Chapter 7, §7.2.4

14 / 18

Slide 15

What does the team need to know?

Scenario: A town wants to reduce road collisions with deer near new housing.

With a partner, write two questions the project team must answer:

1. one question a biological or spatial lens would ask; and
2. one question from a human-dimensions lens.

Then name one field or partner that should be involved before the team chooses an alternative.

Source: Chapter 7, §7.2-7.2.4

15 / 18
